

DW_fp_dp3

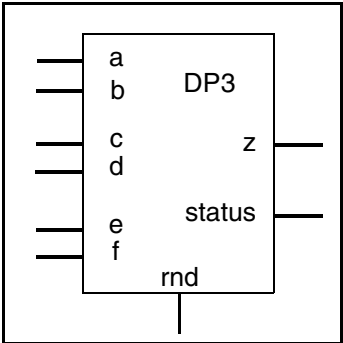
3-Term Floating-Point Dot-product

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Features and Benefits

- The precision is controlled by parameters, and covers formats in the IEEE standard
- Exponents can range from 3 to 31 bits
- Fractional part of the floating-point number can range from 2 to 253 bits
- A parameter controls the use of denormal values
- More accurate than using a combination of basic FP operators.
- Provides a variety of rounding modes.
- DesignWare datapath generator employed for reduced delay and area.

Revision History



Description

DW_fp_dp3 is a floating-point component that computes the dot-product of six floating-point inputs (a, b, c, d, e, and f) to produce a floating-point result $z = a * b + c * d + e * f$, where the symbols (*) and (+) represent floating-point multiplication and floating-point addition, respectively. Therefore, it incorporates three FP multiplications and two FP additions. The accuracy of this component is greater than the accuracy of an implementation using DW_fp_mult components and DW_fp_add components.

Component pins are described in [Table 1-1](#) and configuration parameters are described in [Table 1-2](#).

Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	(sig_width + exp_width + 1) bits	Input	FP input data
b	(sig_width + exp_width + 1) bits	Input	FP input data
c	(sig_width + exp_width + 1) bits	Input	FP input data
d	(sig_width + exp_width + 1) bits	Input	FP input data
e	(sig_width + exp_width + 1) bits	Input	FP input data
f	(sig_width + exp_width + 1) bits	Input	FP input data
rnd	3 bits	Input	Rounding mode; supports all rounding modes described in the Datapath Floating-Point Overview
z	(sig_width + exp_width + 1) bits	Output	$(a * b) + (c * d) + (e * f)$

Table 1-1 Pin Description (Continued)

Pin Name	Width	Direction	Function
status	8 bits	Output	Status flags corresponding to z; for details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i>

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 bits	Word length of fraction field of floating-point numbers a, b, c, d, e, f, and z
exp_width	3 to 31 bits	Word length of biased exponent of floating-point numbers a, b, c, d, e, f, and z
ieee_compliance	0 or 1	Level of support for IEEE 754: 0: No support for IEEE 754 NaNs and denormals; NaNs are considered infinities and denormals are considered zeros 1: Fully compliant with the IEEE 754 standard, including support for NaNs and denormals For more, see IEEE 754 Compatibility in the <i>Datapath Floating-Point Overview</i> .
arch_type	0 or 1	Controls the use of an alternative architecture Default: 0 (previous architecture)

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_DP3_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_dp3_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_dp3.v	Verilog simulation model source code

Table 1-5 Functional Description

a	b	c	d	e	f	status	z ^a
a (FP)	b (FP)	c (FP)	d (FP)	e (FP)	f (FP)	*	a*b + c*d + e*f (FP)

a. The actual value of the result is defined by the rounding mode.

Parameters *ieee_compliance* and *arch_type* control the functionality of this component. Different values of *arch_type* result in slightly different numeric behaviors, but the component is more accurate than the implementation of the same function using basic floating-point multipliers and adders.

When the parameter *arch_type* is set to 0 the component calculates the dot-product function as if the operation was done using infinite precision followed by a single rounding step at the end to generate the final result.

When *arch_type* = 1, a special architecture is used to reduce hardware and the component behaves similar to a network of independent FP operators (network of multipliers and adders), without intermediate rounding. Only one rounding step is performed to compute the output value. In this case, the component produces logic that is not as accurate as when *arch_type* = 0, but it is still more accurate than the network of multipliers and adders. When *arch_type* = 1, the logic created by the component is smaller and faster than when *arch_type* = 0.

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If an invalid rounding mode has been detected on *rnd*, the following message is displayed:

```
WARNING: <instance_path>:  
at time = <timestamp>: Illegal rounding mode.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_dp3_inst is
  generic (
    inst_sig_width : POSITIVE := 23;
    inst_exp_width : POSITIVE := 8;
    inst_ieee_compliance : INTEGER := 0;
    inst_arch_type : INTEGER := 0
  );
  port (
    inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_b : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_c : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_d : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_e : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_f : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    inst_rnd : in std_logic_vector(2 downto 0);
    z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
    status_inst : out std_logic_vector(7 downto 0)
  );
end DW_fp_dp3_inst;
```

architecture inst of DW_fp_dp3_inst is

begin

```
-- Instance of DW_fp_dp3
U1 : DW_fp_dp3
generic map (
  sig_width => inst_sig_width,
  exp_width => inst_exp_width,
  ieee_compliance => inst_ieee_compliance,
  arch_type => inst_arch_type
)
port map (
  a => inst_a,
  b => inst_b,
  c => inst_c,
  d => inst_d,
  e => inst_e,
  f => inst_f,
  rnd => inst_rnd,
  z => z_inst,
  status => status_inst
);
```

```
end inst;

-- pragma translate_off
configuration DW_fp_dp3_inst_cfg_inst of DW_fp_dp3_inst is
  for inst
    end for; -- inst
  end DW_fp_dp3_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_dp3_inst( inst_a, inst_b, inst_c, inst_d, inst_e,
                      inst_f, inst_rnd, z_inst, status_inst );

parameter inst_sig_width = 23;
parameter inst_exp_width = 8;
parameter inst_ieee_compliance = 0;
parameter inst_arch_type = 0;

input [inst_sig_width+inst_exp_width : 0] inst_a;
input [inst_sig_width+inst_exp_width : 0] inst_b;
input [inst_sig_width+inst_exp_width : 0] inst_c;
input [inst_sig_width+inst_exp_width : 0] inst_d;
input [inst_sig_width+inst_exp_width : 0] inst_e;
input [inst_sig_width+inst_exp_width : 0] inst_f;
input [2 : 0] inst_rnd;
output [inst_sig_width+inst_exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_fp_dp3
DW_fp_dp3 #(inst_sig_width, inst_exp_width, inst_ieee_compliance, inst_arch_type)
U1 (
    .a(inst_a),
    .b(inst_b),
    .c(inst_c),
    .d(inst_d),
    .e(inst_e),
    .f(inst_f),
    .rnd(inst_rnd),
    .z(z_inst),
    .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted the description of the <i>ieee_compliance</i> parameter in Table 1-2 on page 2 ■ Added “Suppressing Warning Messages During Verilog Simulation” on page 3 ■ Added this Revision History table and the document links on this page

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